

Windows Sysmon Security Incident Analysis Report

Author

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Role

Future SOC Analyst | Future Cloud Security Engineer

Project Title

SOC Lab – Windows Sysmon Security Incident Analysis

Tools Used

- Microsoft Sysmon
- Windows Event Viewer
- Windows 11
- Command Prompt
- MITRE ATT&CK Framework

Report Date

June 2026

Executive Summary

This project demonstrates the use of Microsoft Sysmon and Windows Event Viewer to perform security monitoring, threat hunting, and incident investigation on a Windows endpoint.

The objective was to collect and analyze Sysmon telemetry, investigate process execution, network connections, file creation activities, image load events, DNS queries, and registry modifications, and map observed behaviors to the MITRE ATT&CK framework.

Throughout the investigation, Sysmon successfully captured detailed endpoint activity, enabling visibility into user actions and operating system processes. Multiple event categories were analyzed to determine whether any indicators of compromise (IOCs) or suspicious activities were present.

The investigation concluded that all observed activities were consistent with legitimate user and system operations, with no evidence of malicious execution, persistence, privilege escalation, or command-and-control communication.

Project Objectives

The primary objectives of this project were:

- Deploy and validate Sysmon monitoring capabilities
 - Analyze Windows security telemetry
 - Investigate process creation activities
 - Monitor network connections
 - Track file creation events
 - Analyze image load activities
 - Examine DNS query events
 - Perform threat hunting using Sysmon logs
 - Map findings to MITRE ATT&CK
 - Produce a SOC-style incident investigation report
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Lab Environment

Component	Description
Operating System	Windows 11
Monitoring Tool	Microsoft Sysmon
Analysis Tool	Windows Event Viewer
Investigation Method	Manual Log Analysis
Framework	MITRE ATT&CK

Sysmon Deployment Verification

Sysmon Operational logs were verified through Event Viewer.

Path:

Applications and Services Logs

→ Microsoft

→ Windows

→ Sysmon
→ Operational

The successful generation of multiple Sysmon event IDs confirmed that endpoint telemetry collection was functioning correctly.

Observed Event Categories:

- Event ID 1 – Process Creation
- Event ID 3 – Network Connection
- Event ID 7 – Image Load
- Event ID 11 – File Creation
- Event ID 13 – Registry Value Set
- Event ID 22 – DNS Query

Investigation Phase 1 – Process Creation Analysis

Event ID 1

Purpose:
Monitor process execution activity.

Observed Activity

Processes identified:

- cmd.exe
- ipconfig.exe
- rundll32.exe

Key Evidence

Command Line:

ipconfig

Parent Process:

cmd.exe

User:

Vasantha\nitin

Security Analysis

The process execution chain demonstrated normal administrative and discovery activity performed by the authenticated user.

No suspicious parent-child relationships were observed.

No evidence of:

- PowerShell abuse
- LOLBins misuse
- Script execution
- Encoded commands

Risk Assessment

Low Risk

Investigation Phase 2 – Network Connection Analysis

Event ID 3

Purpose:

Monitor outbound and inbound network communications.

Observed Activity

Application:

Brave Browser

Protocol:

UDP

Destination Port:

5353

Security Analysis

The connection was identified as Multicast DNS (mDNS) traffic used for local service discovery.

No connections were observed to:

- Known malicious IP addresses
- Suspicious ports
- Command-and-Control infrastructure

Risk Assessment

Low Risk

Investigation Phase 3 – File Creation Analysis

Event ID 11

Purpose:

Track file creation activity.

Observed Activity

File Created:

New Text Document.txt

Process:

Explorer.exe

User:

Vasanthanitin

Security Analysis

Sysmon successfully captured user-generated file creation activity.

The event demonstrates Sysmon's capability to monitor endpoint file system changes.

No suspicious file creation activity was identified.

Risk Assessment

Low Risk

Investigation Phase 4 – Image Load Analysis

Event ID 7

Purpose:

Monitor DLL and module loading activity.

Observed Activity

Process:

taskhostw.exe

Loaded Module:

urlmon.dll

Signature:

Microsoft Windows

Signature Status:

Valid

Security Analysis

The loaded module originated from a trusted Microsoft source.

No indicators of:

- DLL Hijacking
- Malicious DLL Injection
- Unsigned Modules

were detected.

Risk Assessment

Low Risk

Additional Telemetry Analysis

Event ID 13 – Registry Value Set

Purpose:

Monitor registry modifications.

Observed Activity:

Windows system configuration updates.

Assessment:

Legitimate operating system behavior.

Event ID 22 – DNS Query

Purpose:

Monitor domain name resolution.

Observed Activity:

Browser-generated DNS requests.

Assessment:

Expected endpoint communication activity.

MITRE ATT&CK Mapping

Discovery

T1016 – System Network Configuration Discovery

Observed Tool:

ipconfig.exe

Evidence:

User executed network configuration commands.

T1033 – System Owner/User Discovery

Observed Activity:

User account enumeration through command execution.

Execution

T1059 – Command and Scripting Interpreter

Observed Tool:

cmd.exe

Evidence:

Command shell execution detected.

Command and Control

T1071 – Application Layer Protocol

Observed Activity:

Browser network communication.

Evidence:

Network connections recorded by Sysmon Event ID 3.

T1071.004 – DNS

Observed Activity:

DNS query telemetry.

Evidence:

Sysmon Event ID 22.

Collection

T1074 – Data Staged

Observed Activity:

File creation events.

Evidence:

Sysmon Event ID 11.

Defense Evasion

T1574 – Hijack Execution Flow

Observed Monitoring Capability:

DLL image load tracking.

Evidence:

Sysmon Event ID 7.

Threat Hunting Methodology

The investigation followed a structured SOC workflow:

1. Collect Endpoint Telemetry
 2. Identify High-Value Event IDs
 3. Investigate Process Activity
 4. Review Network Communications
 5. Analyze File System Activity
 6. Validate Image Loads
 7. Correlate Events
 8. Map Findings to MITRE ATT&CK
 9. Document Findings
 10. Recommend Monitoring Improvements
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Security Findings Summary

Finding	Severity
Process Creation Activity	Informational
Network Connections	Informational
File Creation Events	Informational
Image Load Activity	Informational
DNS Queries	Informational
Registry Modifications	Informational

No indicators of compromise were identified during the investigation.

Recommendations

Endpoint Monitoring

- Deploy Sysmon across all endpoints
- Regularly update Sysmon configuration rules

Threat Detection

Monitor:

- powershell.exe
- cmd.exe
- rundll32.exe
- mshta.exe
- certutil.exe
- wscript.exe

Network Monitoring

Investigate:

- Unusual outbound connections
- Rare destination ports
- Suspicious DNS requests

SIEM Integration

Forward Sysmon logs into:

- Splunk
- Microsoft Sentinel
- Elastic Security
- QRadar

MITRE-Based Detection

Develop detection rules mapped to:

- Execution
- Persistence
- Discovery
- Command and Control
- Defense Evasion

techniques.

Conclusion

This project successfully demonstrated the use of Windows Sysmon as an endpoint monitoring and threat hunting solution. Through detailed analysis of process creation, network communications, file system activity, image loads, DNS queries, and registry modifications, comprehensive visibility into Windows endpoint behavior was achieved.

The investigation validated Sysmon's effectiveness for SOC operations, incident response, and proactive threat detection. The project also demonstrated practical skills in log analysis, event correlation, security monitoring, MITRE ATT&CK mapping, and incident investigation, aligning with the responsibilities of a Security Operations Center (SOC) Analyst.

As a Future SOC Analyst and Future Cloud Security Engineer, this project strengthened foundational skills in endpoint monitoring, threat detection, security operations, incident analysis, and security event correlation. The knowledge gained is directly applicable to modern SOC environments and cloud security operations, providing a strong foundation for entry-level Security Analyst, SOC Analyst, and Cloud Security roles.